MAM-YOLOv9: A Multiattention Mechanism Network for Methane Emission Facility Detection in High-Resolution Satellite Remote Sensing Images

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***Abstract*— Over 150 countries have signed the Global Methane Pledge, aiming to reduce anthropogenic methane emissions by 30% by 2030. Reducing methane emissions from the energy sector is crucial to achieving this target. The current emission inventories for the energy sector have a spatial resolution of 1 km, suitable for regional-scale methane flux inversion but inadequate for identifying and monitoring point source emissions which is the most important type of anthropogenic methane emissions in the energy sector. To address this issue, we propose a multiattention mechanism, MAM-YOLOv9, for identifying emission facilities in the oil and gas industry, based on YOLOv9. We integrate SimAM and cascaded group attention (CGA) modules into the network, focusing on target objects under complex backgrounds while improving detection accuracy. In addition, we introduce the dynamic convolution module to replace the convolution in the YOLOv9 backbone network, improving computational efficiency and accurate object detection capability. Using submeter-level optical images provided by the high-resolution satellite images, we achieve large-scale monitoring of facility-level emission sources on a regional scale. Experiments demonstrate that our new method achieved SOTA performance, achieving the best results across various metrics compared with the baseline. We also conduct batch detection tasks in Shengli Oilfield, the second-largest oilfield in China, identifying over 38 000 emission facilities. Based on the results, we further compile a facility-level methane emission inventory, which can better serve the global efforts for mitigating methane emissions from the oil and gas industry.**

***Index Terms*— Attention mechanism, fusion network, methane emission facilities, object detection, remote sensing.**

Received 17 October 2024; revised 7 January 2025; accepted 17 February 2025. Date of publication 25 February 2025; date of current version 13 March 2025. This work was supported in part by the National Key Research and Development Program under Grant 2023YFB4104200, in part by the National Natural Science Foundation of China under Grant 42475144, in part by Hubei Provincial Natural Science Foundation under Grant 2023AFB834 and Grant 202CFD015, and in part by the Fundamental Research Funds for the Central Universities under Grant 4106-413000027. *(Corresponding author: Ge Han.)*

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Digital Object Identifier 10.1109/TGRS.2025.3545034

I. Introduction

METHANE is the second-largest anthropogenic greenhouse gas after carbon dioxide, but compared with carbon dioxide, methane is a much more potent greenhouse gas and has contributed to a 0.6 °C rise in global warming since the preindustrial era [1], [2]. Compared with carbon dioxide, methane has a shorter atmospheric lifetime but a much higher global warming potential (GWP), making it more impactful for achieving near-term climate goals [2], [3]. Due to the significant impact of methane emissions on global radiative forcing, reducing methane emissions is considered an effective short-term strategy to mitigate global warming [3]. Therefore, over 150 countries have signed the Global Methane Pledge in the past 4 years, aiming to reduce anthropogenic methane emissions by 30% by 2030 [2]. In the energy sector, particularly in the oil and gas industry, methane emissions primarily occur as leaks [4] rather than organized emissions such as carbon dioxide [1]. Therefore, reducing methane emissions is less costly and can even yield economic benefits beyond climate effects [5]. The existing methane emission inventories have a spatial resolution of 1–10 km, which plays a crucial role in revealing overall regional emissions [6]. However, this resolution is inadequate for identifying superemitters at the point source scale [4], [7]. Recent airborne remote sensing measurements and satellite observations in the Permian Basin of the United States have shown that point source emissions are the primary form of methane emissions in the oil and gas industry, contributing 50%–80% of the total emissions [5], [8], [9]. Current point source imaging satellites (fine pixels < 60 m) such as PRISMA and Landsat-8/9 aim to quantify individual point sources by imaging plumes, while regional flux mapping satellites (high precision $< 1\%$, pixel size 0.1–10 km) such as GOSAT and TROPOMI quantify emissions at the regional scale [10]. Therefore, focusing on methane emission facilities in large-scale oil fields holds significant practical importance for monitoring methane emissions [11], [12]. This approach not only allows for precise identification of the specific

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locations of methane emission facilities but also provides data support for greenhouse gases' emission accounting, helping governments better regulate methane emissions in the oil and gas sector [13], [14]. In this context, compiling methane emission inventories at the facility level becomes a task of significant practical importance. It can help remote sensing scientists better monitor and quantify methane superemitters [15], [16] and guide the design of new methane observation satellites [17], [18].

In recent years, advancements in optical remote sensing technology and satellite communication have significantly improved both the quality and quantity of optical remote sensing images [19]. The successful implementation of the "Gaofen" special project has enabled the use of civil optical satellite data with submeter resolution, allowing for the efficient large-scale collection of Earth observation data. This development opens up the possibility of detecting potential methane emissions from facilities over extensive areas, such as 100 × 100 km. However, there are several challenges in identifying methane emission sources, as illustrated in Fig. 1. The first challenge is that these facilities are often dispersed over large regions, necessitating extensive remote sensing imagery to encompass them. In addition, they may be integrated within dense urban settings, leading to low contrast between the target and its background, which complicates identification. Furthermore, there is a wide variety of methane emission facilities, each with different equipment designs and shapes. For instance, oil tanks vary in type and size, with some small tanks having tops that are only a few square meters, making the identification process even more challenging.

To address the aforementioned issues, we propose a method for identifying methane emission facilities using high-resolution satellite images. This approach integrates attention mechanisms into machine learning models and combines various data processing methods and extraction strategies. Ultimately, we achieve accurate identification of methane emission facilities in oil fields through the fusion of training samples. Our goal is to bridge the data gap concerning methane emission facilities in the oil and gas industry. The key steps can be summarized as follows:

1) *An enhanced YOLOv9 model is proposed*, which incorporates an attention mechanism module to improve the identification of small targets and the accuracy of bounding box recognition. By reducing the number of parameters and floating-point operations (FLOPs) without significantly increasing the parameter count, the model achieves greater adaptability, enhanced representational capacity, and improved accuracy while maintaining detection precision. Computational efficiency and accurate object detection capabilities are further enhanced by replacing the convolutional layers in the YOLOv9 backbone with a dynamic convolution (DyConv) module, allowing the network to focus more effectively on the target terrain.
2) *A fused methane emission facility sample dataset has been constructed* to validate the effectiveness of the proposed method. This dataset comprises over a thousand meticulously labeled optical images of varying sizes, sourced from multiple relevant methane emission facility datasets, as well as a self-created dataset by GF-satellite images. This article reports on the training of an enhanced YOLOv9 model using our proposed dataset, achieving strong performance.
3) *A method for constructing a vector database of methane emission facilities (MEFD-v1.0) is proposed*. This method is based on the identification results of methane emission facilities obtained from the Gaofen-2 (GF-2) and Gaofen-7 (GF-7) satellites. The methane emission facilities associated with oil fields at the basin scale were vectorized, resulting in the creation of a comprehensive vector database.



Fig. 1. Methane emission facilities in high-resolution optical image. (a) Dense oil wells in a vast area. (b) Dense oil tanks in a vast area. (c) Oil tanks mixed in urban buildings. (d) Oil wells mixed in farmland. (e) Oil tanks of different sizes. (f) Oil tanks of different appearance types.

II. Related Work

A. Methods of Methane Emission Facilities Detection

Deep-learning-based object detection algorithms have made significant advancements in both detection accuracy and speed [20]. Traditional optical satellite remote sensing image target detection primarily relies on machine learning and image processing techniques, which include region selection, feature extraction, and classification [21]. With enhanced feature extraction robustness, improved generalization capabilities, and superior learning capacity, deep learning has emerged as the dominant approach in the field of object detection. Methane emission facilities, such as oil tanks and oil wells, possess distinctive visual characteristics that can be

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identified in conjunction with environmental semantic information, making deep learning methods increasingly prevalent for the identification of these facilities.

Zalpour et al. [22] developed a method for detecting oil tanks using deep learning features extracted from high-resolution satellite images, while maintaining controlled false alarm rates. They used an enhanced Faster R-CNN framework during the region-of-interest (ROI) extraction process to minimize both the training and testing time. Their observations indicated that all the oil tanks in high-resolution images appear as circular objects; consequently, they applied a modified circular Hough transform for rapid circle detection to extract these circular tanks. In addition, VGG16 was used to extract features such as shape, color, and surface texture of the oil tanks, with classification achieved through a linear support vector machine (SVM), resulting in the final oil tank detection results. Zhang et al. [23] developed a teacher–student network in which self-supervised learning was informed by optical image data to pretrain an effective feature extractor for synthetic aperture radar (SAR) images. Knowledge distillation was subsequently used to transfer oil tank recognition capabilities from optical images to SAR images, with the optical image data serving as an object-level fine-tuning guide for SAR feature learning. This approach successfully used optical images to enhance the training of a more effective SAR-based oil tank detector. Song et al. [24] concentrated on the two-stage method known as Faster R-CNN, selecting various framework-based Faster R-CNN models to identify oil wells in optical satellite remote sensing images. Although their research does not include comparisons with other methods, it demonstrates commendable accuracy within the experiments conducted.

The multistage algorithms mentioned above provide accuracy; however, the computation time is high. Consequently, they do not offer an advantage in recognition speed for complex images [25]. In the field of object detection, one-stage detection algorithms, exemplified by the YOLO series, are playing an increasingly significant role. One-stage object detection methods use a regression-based approach to address the problem of object bounding box localization without the need to generate candidate regions in advance. Instead, the object's position and category information are directly regressed through the network [26]. As a result, one-stage detection methods generally provide faster detection speeds while continuously improving accuracy. These methods have also been widely applied in the identification of methane emission facilities.

Xu et al. [27] proposed an enhanced EfficientDet network module for the detection of oil tanks. By incorporating an attention mechanism and a novel myNet module into the EfficientDet model, they mitigated the effects of noise and emphasized the features of small oil tank targets, thereby improving the model's ability to capture these small targets. For model training, they used a multiclass focal loss (MFL) method to enhance training effectiveness. Li et al. [28] proposed an enhanced MSA-YOLO algorithm based on YOLOv4 for real-time oil tank detection. They pruned the multiscale recognition module within YOLOv4's network architecture, thereby simplifying the model while preserving its efficiency. In addition, they used the k-means++ clustering algorithm for the initial selection of anchor boxes and used an optimized CIoU-NMS to accelerate model convergence, resulting in improved inference speed and accuracy. He et al. [29] integrated the residual channel attention network (RCAN) with YOLOv4, standardizing the spatial resolution of various images to effectively detect oil and gas well sites from multisource sensor images. However, this approach is limited to detecting based solely on the shape features of the well sites, leading to relatively low accuracy. Zhang et al. [30] investigated a deep-learning-based method for automatic oil well detection from remote sensing images under occlusion conditions. They combined YOLOv5s segmentation with the class activation mapping (CAM) method and the nonmaximum weighted detection (NWD) method for detecting oil wells, demonstrating that this combination can enhance detection accuracy in various blockage scenarios. The results are presented in the form of instance segmentation.

The aforementioned methods demonstrate that the YOLO series has been extensively used for detecting methane emission facilities in oil fields. While various methods can yield satisfactory results in specific scenarios, these models generally lag in performance compared with the more recent versions of YOLO [31]. Detecting methane emission facilities on a large scale necessitates more precise models capable of managing the complexities present in remote sensing images. Consequently, to enhance the identification of boundaries and detailed information regarding methane emission facilities and to improve detection accuracy, YOLOv9 has emerged as a promising option among the mainstream YOLO series models.

B. Related Datasets

An effective deep learning model relies on large-scale, high-quality datasets. Both the size of the dataset and the quality of the annotated target categories significantly influence the performance and practical effectiveness of deep learning models. High-quality annotations ensure that the model learns accurate features, while a sufficiently large dataset enables the model to generalize more effectively across various scenarios and applications. The public datasets related to methane emission facilities that we thoroughly examined are presented in Table I, which includes the sources of their imagery, spatial resolutions, and sample types.

AID is a large-scale aerial image dataset comprising 10 000 images across 30 categories, with each image sized at 600 × 600 pixels, including 360 images of storage tanks. DIOR is another extensive benchmark dataset containing a total of 23 463 images from 20 categories, with an image size of 800 × 800 pixels. In addition, 200 high-quality tank images have been filtered from this dataset. NWPU VHR-10 is a dataset featuring ten categories, which includes 25 images of storage tanks. While the exact image sizes are variable, they are relatively similar, with an average size of approximately 1152 × 769 pixels. NWPU-RESISC45 consists of 21 types of scenes used for scene classification, among which there are 424 optical images of storage tanks, albeit with a smaller

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TABLE I
REFERENCE DATASET

Dataset	Image Source	Spatial Resolution(m)	Type
AID [32]	Google Earth	0.5 ~ 8	Storage Tank
DIOR [28]	Google Earth	0.5 ~ 30	Storage Tank
NWPU VHR-10 [33]	Google Earth	0.5 ~ 2	Storage Tank
NWPU-RESISC45 [34]	Google Earth	0.2 ~ 30	Storage Tank
LEVIR [35]	Google Earth	0.2 ~ 1	Storage Tank
NEPU-OWod-3.0 [30]	Google Earth	0.48	Oil well
OWDM [36]	Google Earth	0.26	Oil well
NEPU-OWOD V1.0 [37]	Google Earth	0.4	Oil well
RSOD-Dataset [38]	Google Earth、MAP WORLD	0.3 ~ 1	Oil Tank

image size of only 256 × 256 pixels. The LEVIR dataset contains a substantial number of high-resolution images, each measuring 800 × 600 pixels, and includes only three types of targets, comprising 109 images of oil tank types. The oil tanks in the datasets mentioned above were not classified separately and were instead mixed with storage tanks, making it challenging to distinguish between them. However, there are several oil tank and oil well datasets that can be used directly, such as NEPU-OWod-3.0, NEPU-OWOD V1.0, OWDM, and the RSOD-Dataset. NEPU-OWod-3.0 contains hundreds of images of Chinese cities and oilfields, as well as 130 images of California, all with a resolution of 512 × 512 pixels. NEPU-OWOD V1.0 comprises a total of 432 oil well sample images with resolutions of 768 × 768 pixels, 768 × 1024 pixels, and 1024 × 1024 pixels, all of which are high-resolution. OWDM integrates a dataset with an image size of 300 × 300 pixels and uses data augmentation techniques to expand the dataset, ultimately resulting in 5895 images along with data annotations. The RSOD-Dataset is a remote sensing collection consisting of only four categories, with oil tanks classified separately, totaling 1586 labeled oil tanks across 165 images.

Although the datasets previously introduced exhibit high image quality and have been extensively used, several issues remain. First, the dataset's image sources are overly homogeneous, with nearly all the images derived from Google Earth. This lack of diversity in the training samples compared with the target images may result in a reduction in detection accuracy for high-resolution images. Second, the majority of categories within the aforementioned datasets do not specifically pertain to methane emission facilities. Consequently, it is imperative to develop a specialized dataset focused on methane emission facilities to facilitate their extraction from images. Therefore, this article proposes the incorporation of two satellites with superior spatial resolution from the high-resolution series, namely, GF-2 and GF-7, to augment the existing remote sensing dataset and collaboratively construct a dedicated dataset for methane emission facilities.

GF-2 represents China's inaugural independently developed civilian optical remote sensing satellite, featuring a spatial resolution superior to 1 m. It is equipped with two cameras: one panchromatic and one multispectral, with resolutions of 1 and 4 m, respectively [39]. The satellite possesses a revisit period of 69 days, which facilitates extensive coverage of regions situated between 80° north and south latitudes. Throughout a single revisit cycle, the satellite completes 1021 orbits around the Earth, thereby achieving comprehensive global coverage. With a side swing capability of ±23°, it is capable of revisiting any location on the planet within a span of five days. The GF-2 satellite provides Level-1 radiometrically corrected products, as well as PMS data products, which include 0.8-m panchromatic and 3.2-m multispectral imagery. The GF-7 satellite is outfitted with a dual linear array stereo camera and a laser altimeter, representing a significant advancement in submeter-level stereo mapping camera technology. It is capable of capturing high spatial resolution optical stereo observation data alongside high-precision laser altimetry data. The dual linear array stereo camera on the GF-7 effectively acquires panchromatic stereo images with a resolution exceeding 0.8 m and a swath width of 20 km, in addition to multispectral images with a resolution of 3.2 m [39].

Overall, these two high-resolution satellites demonstrate commendable performance regarding various performance parameters and image coverage, thereby fulfilling the requirements for the development of methane emission datasets.

III. METHODS

YOLOv9 [31], which was released in February 2024, represents the latest iteration of the YOLO series. This version incorporates innovative technologies, including programmable gradient information (PGI) and generalized efficient layer aggregation network (GELAN). In this article, we present an enhanced model derived from YOLOv9, referred to as MAM-YOLOv9. The architecture of the MAM-YOLOv9 network is illustrated in Fig. 2.

The MAM-YOLOv9 model structure consists of three main parts: the backbone, the head, and the detection layers. Layers 1–12 serve as the backbone, while P3–P5 are the detection layers. Specifically, P3 features a map size of 80 × 80 pixels, designed for detecting targets with a size of 8 × 8 pixels or larger; P4 has a feature map size of 40 × 40 pixels, suitable for detecting targets of at least 16 × 16 pixels; and P5 offers a map size of 20 × 20 pixels, which is optimized for detecting targets of at least 32 × 32 pixels. The remaining layers are designated as head layers.

To enhance the backbone, we introduced the cascaded group attention (CGA) to the model and positioned three CGA modules before the three RepNSPELAn4 modules.

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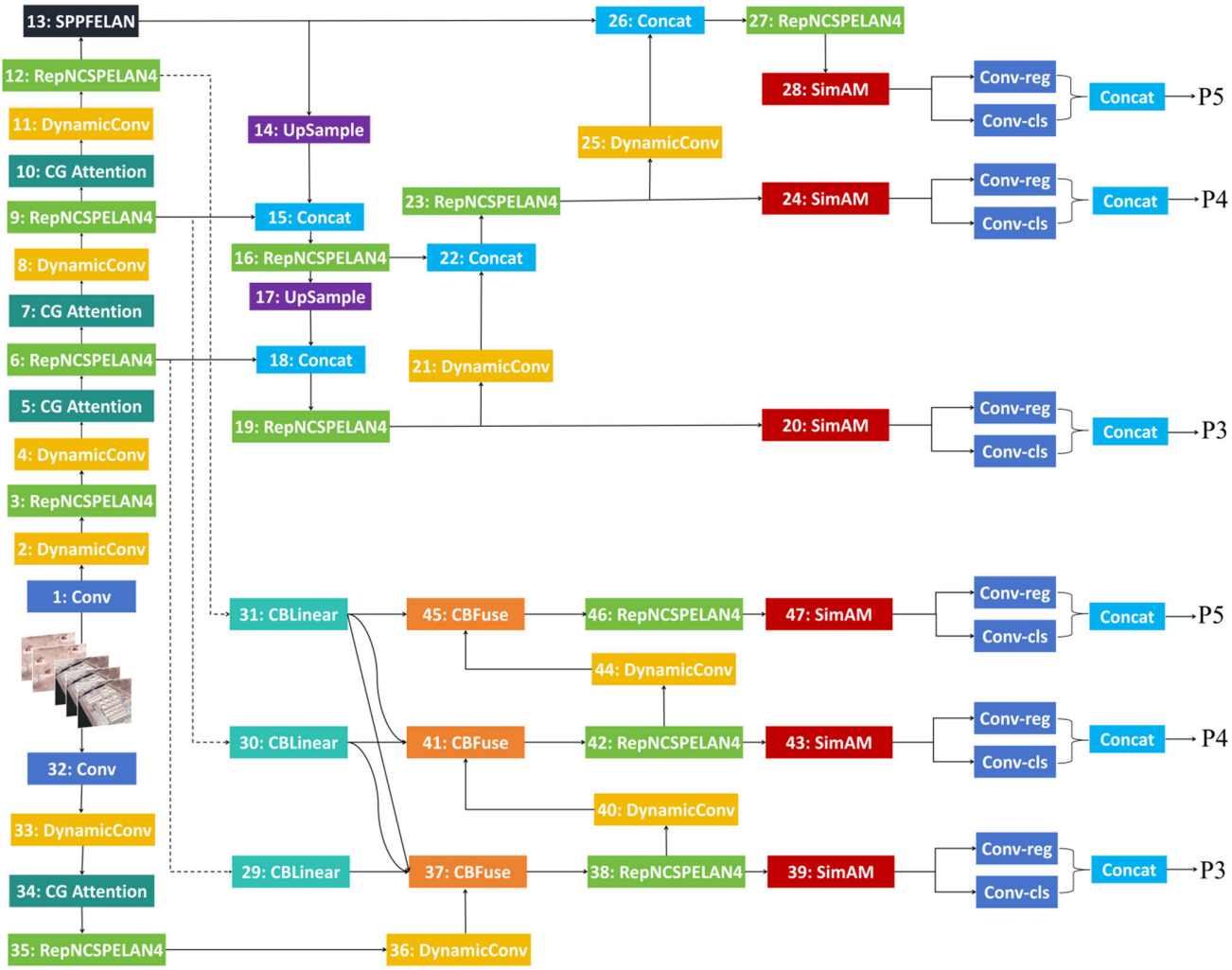


Fig. 2. Model structure of MAM-YOLOv9.

RepNSPELAn4 is a composite network comprising RepConv [40], CSPNet [41], and GELAN, playing a crucial role in feature connection within the model. The RepNSPELAn4 layers in backbone connect two Concat layers and three CBLinear layers in the head part, making it pivotal for feature transfer. Inserting the CGA module in front of the RepNSPELAn4 layer aims to enhance the diversity of the feature maps and improve the overall model performance.

In addition, we made improvements to the head part of the model by incorporating SimAM modules into layers 20, 24, 28, 39, 43, and 47, all of which are positioned before the six detection layers. This application of SimAM enhances the model's ability to concentrate on relevant target objects using the selective attention regions extracted by SimAM.

Finally, we replaced the Conv layers throughout both the backbone and the head with DyConv layers. This modification allows for the combination of multiple convolution kernels without increasing the network's width or depth, thereby augmenting the model's expressive capabilities. Given that YOLOv9 is a relatively new model, we have opted to optimize it within the existing framework rather than implement significant structural changes. Below, we will provide a more detailed explanation of the methods and functions of each module.

A. SimAM

SimAM [42] represents a conceptually straightforward yet highly effective attention module. It is grounded in fundamental theories of neuroscience and is characterized as a lightweight module that is straightforward to implement, making it compatible with YOLOv9. In contrast to the majority of existing channel and spatial attention modules, SimAM is capable of inferring 3-D attention weights for feature maps within the layers without introducing additional parameters to the original network. The structural diagram of SimAM is shown in Fig. 3.

Drawing upon the well-established theory of spatial compression, SimAM formulates an energy function to facilitate this process, defining the energy function for each individual neuron. The specific formula is presented as follows:

$$e_t(\omega_t, b_t, y, x_i) = \frac{1}{M-1}\sum_{i=1}^{M-1}(-1-(\omega_t x_i + b_t))^2 + (1-(\omega_t t + b_t))^2 + \lambda\omega_t^2 \quad (1)$$

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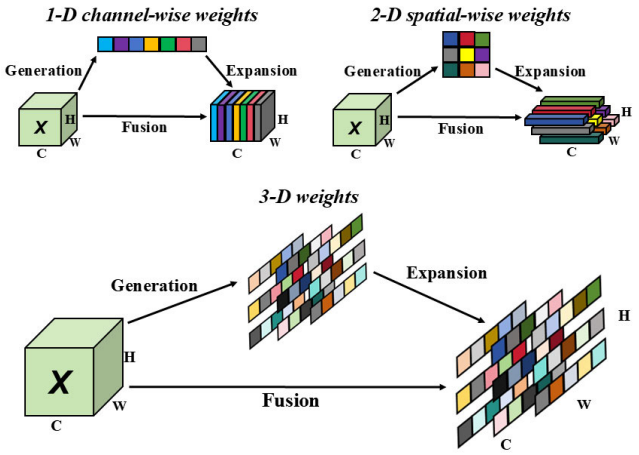


Fig. 3. Schematic of SimAM. Each subgraph represents an attention map in different dimensions; each color represents the use of a single scalar for each point on each channel, spatial position, or feature.

where t is the target neuron, and x_i is other neuron in a single channel of the input feature. i is the index over the spatial dimension and $M = H \times W$ is the number of neurons on that channel. ω_t and b_t are the weight and bias of the transform, respectively. y denotes binary labels, which are used in mathematical formulations to facilitate calculations using 1 and -1 to signify two distinct states or features. This labeling approach enhances the computational efficiency of the model and effectively delineates the differences between neurons. λ is identified as a hyperparameter that regulates the intensity of regularization.

The authors propose a rapid closed-form solution, which will not be elaborated upon in this context. Their methodologies are predicated on the assumption that all the pixels within a single channel adhere to the same statistical distribution. Based on this assumption, the mean μ_t and variance σ_t^2 of all the neurons can be calculated and subsequently applied to all the neurons within this channel. This approach can substantially decrease computational expenses and eliminate the need for iterative calculations of μ and σ for each position. Ultimately, the minimal energy can be determined using the following equation:

$$e_t^* = \frac{4\left(\hat{\sigma}^2 + \lambda\right)}{\left(t - \hat{\mu}\right)^2 + 2\hat{\sigma}^2 + 2\lambda} \tag{2}$$

where the energy e_t^* is lower, the difference between neurons t and the surrounding neurons is higher, and the importance of t is higher. Thus, the importance of neurons can be calculated by $1/e_t^*$.

Therefore, by optimizing the energy function to ascertain the attention significance of each neuron within the feature map, SimAM facilitates the extraction of attention regions through the assessment of neuron importance. In oil field images that encompass numerous emission facilities, extensive target areas frequently exhibit intricate and varied geographical elements. The application of SimAM enables the model to concentrate on pertinent target objects by extracting attention regions, thereby assisting the model in managing complex information. Furthermore, SimAM is a parameter-free attention module, and it enhances the model's feature extraction capabilities without the introduction of additional parameters.

B. Cascaded Group Attention

CGA [43] is an attention module designed for visual transformers. This module facilitates the partitioning of complete feature sets for each attention head, thereby explicitly decomposing the attention computations across the heads. By supplying distinct feature partitions to each head, CGA enhances the diversity of the attention maps. Furthermore, the implementation of cascaded attention heads contributes to the deepening of the network, which in turn improves the model's computational efficiency without the necessity of introducing additional parameters. Given that the attention map computations in each head use smaller channel dimensions, this approach results in only a minimal latency overhead. The computational process of CGA can be articulated as follows:

$$\tilde{X}_{ij} = \text{Attn}\left(X_{ij}W_{ij}^Q, X_{ij}W_{ij}^K, X_{ij}W_{ij}^V\right) \tag{3}$$

$$\tilde{X}_{i+1} = \text{Concat}\left[\tilde{X}_{ij}\right]_{j=1:h} W_i^P \tag{4}$$

where the jth head computes the self-attention over X_{ij}, which is the jth split of the input feature X_i, i.e., $X_i = [X_{i1}, X_{i2}, \ldots, X_{ih}]$ $(1 \le j \le h)$. h is the total number of heads. W_{ij}^Q, W_{ij}^K, and W_{ij}^V are the projection layers mapping the input feature split into different subspaces, and W_i^P is a linear layer that projects the concatenated output features back to the dimension consistent with the input.

This method of segmenting the head significantly reduces computational complexity. For instance, in the experiment presented in this article, the default input is an RGB image with dimensions of $640 \times 640 \times 3$ pixels. Without the head segmentation method, the input sequence consists of $N = 640 \times 640 = 409\,600$ pixels, with each pixel containing three channels. The size of the QKV feature dimension is denoted as d, which is a learnable hyperparameter. In this analysis, we assume $d = 256$, resulting in a QKV matrix size of $N \times d$, and the total number of parameters required for calculating QKV is $3 \times N \times d = 314\,572\,800$, representing a substantial computational burden. Furthermore, the computational complexity is $O(N^2) = O(409\,600^2)$. In contrast, the CGA method divides the image into h equally sized heads. Assuming $h = 16$, the size of each head is $N_1 = 40 \times 40 = 1600$, allowing the pixels in each region to be treated as separate local attention calculation units during computation. Consequently, the new QKV feature dimension size d_1 is set to 64, leading to a QKV matrix size of $N_1 \times d_1$. The parameter count for each head is then calculated as $3 \times N_1 \times d_1 = 307\,200$. For all the heads combined, the total parameter count is $16 \times 307\,200 = 4\,915\,200$, which represents a relative reduction of approximately 98.4% in the number of parameters. In addition, the computational complexity for each head in the CGA method is $O(N_1^2) = O(1600^2)$, which significantly decreases the computational complexity associated with each head.

CGA enhances computational efficiency using feature partitions rather than using the complete set of features for each attention head, thereby reducing the computational load

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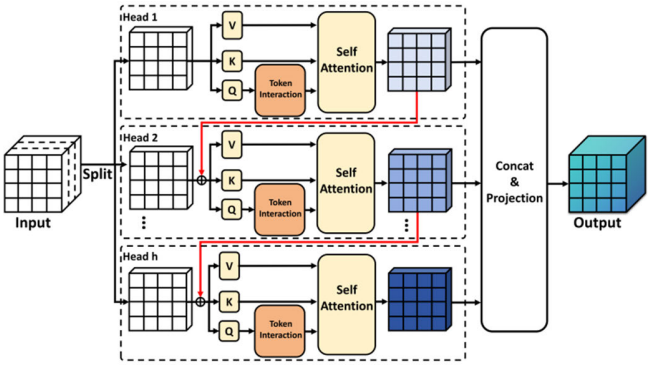


Fig. 4. Schematic of CGA. The red line represents adding the output of each header to the subsequent headers to jointly calculate self-attention. Besides, another token interaction layer is applied after Q projection, which enables self-attention to jointly capture local and global relations and further enhances feature representation.

associated with each head. Furthermore, CGA augments its capacity by learning more complex feature mappings through the integration of additional layers. The structure is illustrated in Fig. 4. The attention maps for each head are computed in a cascaded fashion, where the output of each head is incorporated into the input of the subsequent head

$$X'_{ij} = X_{ij} + \tilde{X}_{i(j-1)}, \quad 1 < j \leq h \tag{5}$$

where X'_{ij} is the addition of the jth input split X_{ij} and head output $\tilde{X}_{i(j-1)}$ calculated by (3) and (4). It replaces X_{ij} to serve as the new input feature for the jth head when calculating the self-attention. Each head computes its self-attention map, after which the outputs from all the heads are concatenated and subsequently projected back to the original input dimension via a linear layer. This approach not only enhances the model's capacity but also promotes feature diversity and progressively refines the representation of features.

We attempted to improve the CGA framework by not only the incorporation of the output from each subhead X_{ij} into the input of the next head X'_{ij} but also the inclusion of the output from the immediately preceding head $X_{i(j-1)}$. However, the results showed little success. In summary, CGA enhances the diversity of attention, enabling models to more effectively concentrate on emission facility targets within complex backgrounds, particularly in scenes that contain a mixture of dense features. Furthermore, the computational methodology used by CGA does not impose an excessive burden on the model; rather, it enhances the model's capacity, thereby contributing to improved robustness.

C. Dynamic Convolution

Although YOLOv9 has deepened its network depth compared with its previous versions, it still belongs to the category of lightweight networks. However, compared with high-performance deep networks, lightweight networks have limitations on the depth and channels of the network, resulting in decreased performance, limited feature expression ability, and decreased recognition performance. In DyConv [44], the convolution kernels are related to the input, enabling different data to generate distinct convolution operations. This mechanism enhances the model's expressive capability without increasing the network's width or depth. To balance performance and computational efficiency, multiple convolutional kernels are fused, further improving the model's representational power. Since this study uses multisource data samples, the input data to the model vary dynamically. Accordingly, we replaced the original convolution module in YOLOv9 with DyConv, where the kernel weights adapt dynamically to the input data. By adjusting the weights in real-time, the model better captures diverse input features, significantly enhancing its expressive capacity.

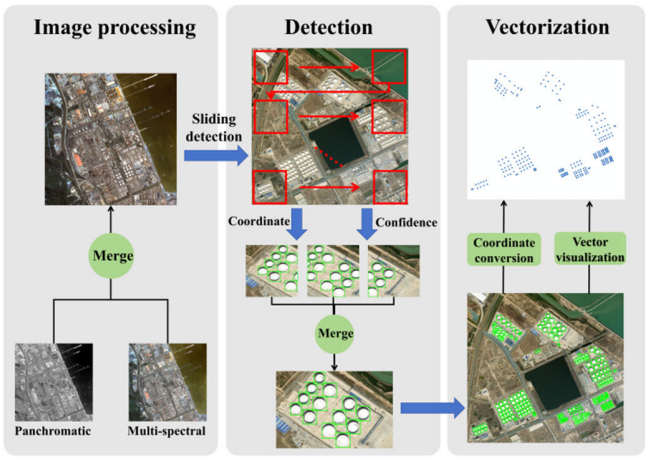


Fig. 5. Proposed method for constructing vector database. First, the fused high-resolution images are fed as data sources into the network for detection. Then optimize the detection results based on the coordinates and confidence of the recognition results. Finally, a vector database is established through coordinate transformation and vector visualization.

D. Construction of Vector Database

Duan et al. [45] proposed a method leveraging multiple remote sensing platforms for collaborative processing to detect targets obstructed by clouds and mist. Inspired by this approach, we use GF-2 and GF-7 imagery as data sources to detect methane emission facilities. Initially, multispectral and panchromatic images are fused. The fused images retain both high spatial resolution and rich spectral information, offering visual advantages over standalone panchromatic or multispectral images. These combined images not only provide clearer details but also richer color and contrast, facilitating more accurate identification of methane emission facilities. The specific process diagram is shown in Fig. 5.

Due to the large size of high-resolution images, they cannot be directly processed by convolutional neural networks [46]. To address this, we use a sliding window mechanism to traverse the entire image, effectively handling size constraints. However, this process may cause the same object to appear in multiple sliding windows, leading to duplicate detections and errors. To mitigate these issues, a certain degree of overlap is introduced between sliding windows. This reduces both missed detection and false detection rates. Although sliding window detection can be computationally inefficient, the proposed MAM-YOLOv9 compensates for this shortcoming with its lightweight design.

During the sliding process, the relative position of each window within the overall image is recorded. Based on recognition

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results, the positional information of the target device relative to the sliding window is also captured. The relative position format is defined as x, y, width, height, and conf, where x and y represent the normalized center coordinates of the target, width and height are the normalized dimensions, and conf indicates the confidence level.

Merging detection boxes refers to merging the detection results of overlapping areas. When the same target has multiple sliding window detection results, and the overlap is greater than the set threshold, the detection boxes with low confidence will be removed, and the remaining detection boxes are merged to obtain the final detection result. Finally, by combining the relative position coordinates of the sliding window and the relative coordinates of the target within the sliding window, the true geographic latitude and longitude coordinates of the target can be calculated using the following equations:

$$X = \text{Lon1} \pm |\text{Lon2} - \text{Lon1}| \frac{X1 + x \times \text{Slide_Width}}{\text{GF_Width}} \tag{6}$$

$$Y = \text{Lat1} \mp |\text{Lat2} - \text{Lat1}| \frac{Y1 + y \times \text{Slide_Height}}{\text{GF_Height}}. \tag{7}$$

Among them, (X, Y) is the true geographic coordinate longitude and latitude of the target. (Lon1, Lat1) is the geographic coordinate longitude and latitude of the upper left corner of the image. (Lon2, Lat2) is the geographic coordinate longitude and latitude of the lower right corner of the image. (x, y) are the normalized center coordinates of the target relative to the sliding window. Slide_Width and Slide_Height represent the length and width of the sliding window, and GF_Width and GF_Height represent the length and width of the image, respectively. After calculation, the relative image coordinates of the target can ultimately be converted into real latitude and longitude coordinates.

Finally, a methane emission facility vector database can be established.

IV. Experiment

A. Study Area

The research area in this work is the Shengli Oilfield and its surrounding areas in Dongying, Shandong Province, China. This region is located in the Bohai Bay Basin and has abundant oil and natural gas resources, making it the second largest oilfield in China. Up to now, Shengli Oilfield has discovered many billion-ton oil fields in major depressions, with proven oil reserves exceeding one billion tons and considerable natural gas reserves. As for China, the development of this oil field plays an important role in her energy safety. The images of Shengli Oilfield come from GF-2 and GF-7 satellites. Due to the large size of the study area, high-resolution images on different tracks are required to fully cover the area. Therefore, there will be intersections between high-resolution images, and images with overlapping areas will be overlaid to ultimately output a complete fused image and convert it into PNG format.

B. Dataset and Processing

This article constructs a fused methane emission facility sample dataset to validate the effectiveness of the proposed method. The dataset contains over a thousand carefully labeled optical images of varying sizes, with samples sourced from various related methane emission facility datasets in Table I, as well as self-produced methane emission facility data based on high-resolution satellite imagery. Since most of the sample data in the external datasets are classified as "storage tanks" (including oil tanks, water tanks, grain silos, etc.) rather than just oil tanks, and because different types of tanks exhibit distinct image features, oil tanks can be carefully selected based on expert knowledge. The oil well samples in the external datasets are all categorized as "oil well" and do not require additional filtering. The labeled sample images are shown in Fig. 6.

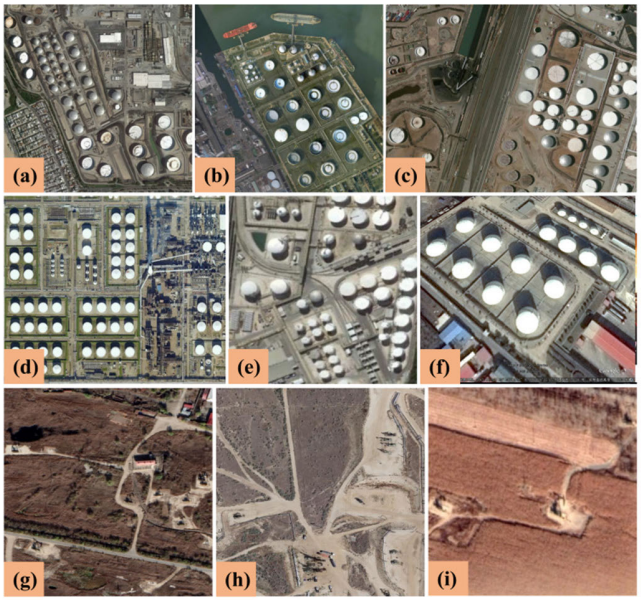


Fig. 6. Sample of multisource methane emission facilities. (a) AID. (b) DIOR. (c) LEVIR. (d) NWPU VHR-10. (e) NWPU-RESISC45. (f) NWPU dataset. (g) NEPU-OWOD V1.0. (h) NEPU-OWod-3.0. (i) OWDM.

This article uses visual interpretation methods to conduct pixel-level annotations of methane facility dense distribution areas on 10 GF-2 and GF-7 remote sensing images, with mutual validation to ensure annotation accuracy. A total of 200 sample images, each 640 × 640 pixels, were cropped from high-resolution images, including 100 oil tank samples and 100 oil well samples.

Along with the organized external dataset samples, the final methane emission facility dataset compiled in this study includes 1150 oil tank samples and 850 oil well samples, with a total of 10 695 oil tank annotations and 2295 oil well annotations. On average, each oil tank sample contains 9.3 annotations, while each oil well sample has 2.7 annotations, essentially meeting the data volume requirements for training a methane emission facility identification task.

As shown in Fig. 7, due to the height of the oil tank, it will have different shadows at different angles of sunlight, and the oil well pumping units will have different directional distributions in the image. Therefore, to better identify oil well pumping units distributed in different directions and oil tanks under different directions of solar radiation, we rotated

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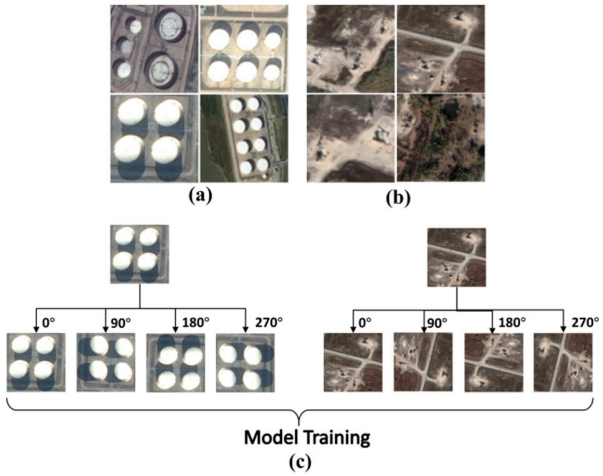


Fig. 7. Characteristics of sample data and data augmentation methods. (a) Oil tanks in different directions. (b) Oil wells in different directions. (c) Select and enhance oil tank and oil well samples in four directions before entering the model for training.

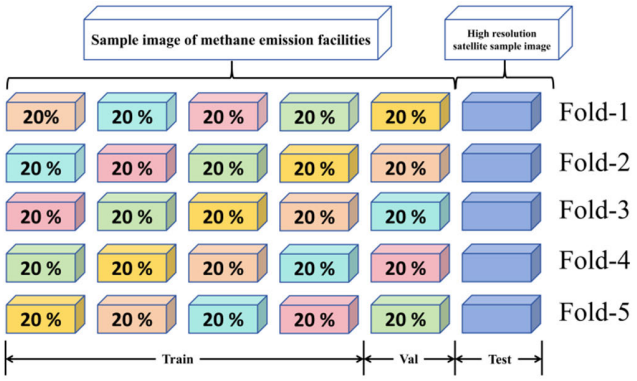


Fig. 8. Fivefold cross-validation. The parts with the same color represent the same sample images.

the training samples at angles of 0°, 90°, 180°, and 270°. By rotating the training box to increase input data, we can improve detection accuracy and better detect methane emission facilities in oil fields from different directions.

As part of the model training process, this study used the default data augmentation techniques of the YOLOv9 model, including HSV tone enhancement, image translation, image scaling, image left and right flipping, and image tiling.

This technology helps expand the amount of data used in the training process and promotes adaptation to sharp changes in object size in images. It enhances the model to perform better object detection in images, reduces the risk of overfitting, and enhances the model to perform object detection in different environments. It is worth emphasizing that due to the fact that the images in the dataset come from different regions, there are differences in the morphology, size, and background of methane emission facilities. Therefore, to avoid the randomness of model performance caused by the test dataset and fully use all the data, we applied k-fold cross-validation (KFCV) method in the training and evaluation of each experiment. Our study divided the data into five parts for cross-validation, as shown in Fig. 8. Divide 80% of the dataset into the training set and 20% into the validation set. The partitioning of the training and validation sets is performed randomly to maintain their diversity and representativeness. Due to the fact that the final training model needs to be oriented toward high-resolution series satellite images, high-resolution satellite image samples were selected for the test set. Take the average of the results of each model to obtain the average result of that model. The model with the best average result among the five models trained is the optimal model.

C. Implementation Details

The experiment in this article used an NVIDIA GeForce RTX 4090 GPU with 24 GB of memory and Python and PyTorch frameworks. Before training our model, we used YOLOv9 model weights pretrained on the MS COCO [47] dataset. During the training process, we use the SGD optimizer to train the model with an initial learning rate of 0.001, momentum of 0.93, weight decay rate of 0.0005, and epoch of 1000. Due to the memory limitation of a single GPU, a batch size of 16 was used to train the model.

Model evaluation is crucial for determining model performance and compatibility with research objectives. These include those related to the accuracy, speed, and efficiency of the model. The commonly used basic evaluation indicators are precision and recall. The calculation of accuracy and recall rate is shown in (8) and (9). TP represents true positives, referring to the situation where an object is correctly predicted to be a positive sample. FP represents false positives, which refers to detecting an invalid object as a positive sample. FN represents false negatives, which means that a valid object is detected as a negative sample [48]

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{8}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{9}$$

In addition to accuracy and recall, there are other evaluation metrics more commonly used for object detection, namely, average precision (AP) and mean AP (mAP). The AP measurement model searches for relevant objects and ignores irrelevant objects. AP is calculated by drawing a precision recall curve and calculating the area under the curve using (10). AP_i represents the AP value of the ith category, and k is the number of categories. mAP has been applied in many other object detection studies to demonstrate the accuracy and recall trends of various training models [49]. mAP is the average AP of all the object classes found by the model. mAP provides a more comprehensive picture of model performance as it measures the accuracy of all the categories of objects, not just one category. The equation is shown in (11)

$$\text{AP} = \int_0^1 \text{Precision}(r)dr \tag{10}$$

$$\text{mAP} = \frac{\sum_{i=1}^{K} \text{AP}_i}{k}. \tag{11}$$

In addition to evaluation metrics related to accuracy, there are also evaluation metrics related to model speed and efficiency. For example, GFLOPs measures the number of FLOPs

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TABLE II
COMPARATIVE EXPERIMENTAL RESULTS

Model	Precision	Recall	mAP@0.5	mAP@0.5~0.95	Params(M)	FLOPs(G)	Speed(ms)
Faster R-CNN [50]	0.8039	0.7891	0.8207	0.4482	138.4	185.1	253.97
SSD [51]	0.7934	0.7675	0.8010	0.4163	32.2	**39.5**	**41.12**
YOLOv7 [52]	0.8973	0.8879	0.8651	0.6151	36.5	103.2	114.34
YOLOv8-X [53]	0.9245	0.9047	0.9292	0.6543	68.2	257.8	179.27
YOLOv10-X [54]	0.9588	0.9298	0.9684	0.7383	**29.5**	160.4	58.84
MAM-YOLOv9	**0.9734**	**0.9666**	**0.9894**	**0.7980**	35.1	187.2	66.72

that a model can perform in 1 s, while speed measures the time required for the model to process a single image. In addition, there are other evaluation indicators, such as the number of model parameters Params, which refers to the number of parameters used by the model to learn patterns from the training data. The more parameters used, the more complex the model becomes.

D. Comparative Experiment

To evaluate the performance of the proposed algorithm MAM-YOLOv9 and other strong models on high-resolution images, we have specifically selected the current mainstream object recognition algorithms Faster R-CNN, SSD, YOLOv7, YOLOv8-X, and YOLOv10-X for comparative experiments. The quantitative results are shown in Table II. This study compared the model size (parameters and FLOPs), accuracy (accuracy, recall rate, 50% AP (mAP@0.5), 50%–95% AP (mAP@0.5–0.9), and inference time. The bold parts in the table represent the optimal results for this metric.

The experimental results show that SSD performs the worst in the detection of methane emission facilities, but considering the low FLOPs and speed, the low detection accuracy is understandable. Faster R-CNN is a multistage method that has not achieved good results in detection, only showing slightly better performance than SSD, but with the largest number of model parameters among all the models. YOLOv7 is a previous version of YOLOv9, which is an improvement on YOLOv7. YOLOv7 performs well in terms of model parameters, but the performance on mAP@0.5 is still not excellent enough. And two relatively new YOLO series algorithms, YOLOv8-X and YOLOv10-X, have shown good and similar performance in precision, recall, and mAP@0.5. In terms of indicators, YOLOv8-X is relatively close to YOLOv10-X, but YOLOv10-X has a more significant improvement in model parameters, FLOP speed, and inference time. The experimental results show that the improved MAM-YOLOv9 model proposed in this article has best accuracy and recall rate. The mAP@0.5 and mAP@0.5–0.9 are also superior to other models. The MAM-YOLOv9 is the only model among them that scores over 95% in terms of precision, recall, and mAP@0.5. Compared with other models, mAP@0.5–0.9 also shows significant improvement. Although MAM-YOLOv9 does not achieve optimal model size and detection speed, it uses DyConv to improve computational efficiency and object detection accuracy and adds attention modules to the backbone network to enhance useful information and suppress background noise around the target. This enables methane emission facilities of different sizes to be better detected, still achieving optimal model performance on the basis of relatively small model size and relatively fast inference time.

To further compare MAM-YOLOv9 with other methods, this article selected some oil tank and oil well areas in the research area of Shengli Oilfield as samples and used different methods to detect methane emission facilities and compare the detection results. The comparison of detection results from different methods is shown in Fig. 9. Positive objects detected correctly, missing objects, and negative objects detected incorrectly are represented in green, red, and yellow, respectively. The sample area includes oil tanks of various colors and surface structures, as well as oil wells in different directions. According to the experimental results, the accuracy of the model is directly proportional to the detection results.

In Table III, since the false detection rates of several methods are all controlled within 3%, we only counted the positive detection rate and missed detection rate and presented them in percentage stacking. Due to unfavorable size and background noise, there are still false positives and false negatives in the area. Based on a detailed analysis of the experimental results, it can be concluded that MAM-YOLOv9, compared with other algorithms, detected a more sufficient number of multiscale target objects and exhibited the lowest false-positive and false-negative rates. In addition to comparing the detection performance of sample images of specific sizes, it is also necessary to explore the recognition performance of the model on a large range of high-resolution images, that is, the impact of selecting sliding windows of different sizes on the model's performance and seeking the optimal window size. In this section, we compare the recognition performance of MAM-YOLOv9 using three different sizes of sliding windows. The sizes of the windows are 300 × 300 pixels, 480 × 480 pixels, 640 × 640 pixels, and 720 × 720 pixels, respectively. The selection of these three sizes is based on the size range of the reference training samples. The detection data are sourced from GF-7 satellite imagery, and the actual number of methane emission facilities obtained through manual visual interpretation is 2698, which is used as the true value to compare

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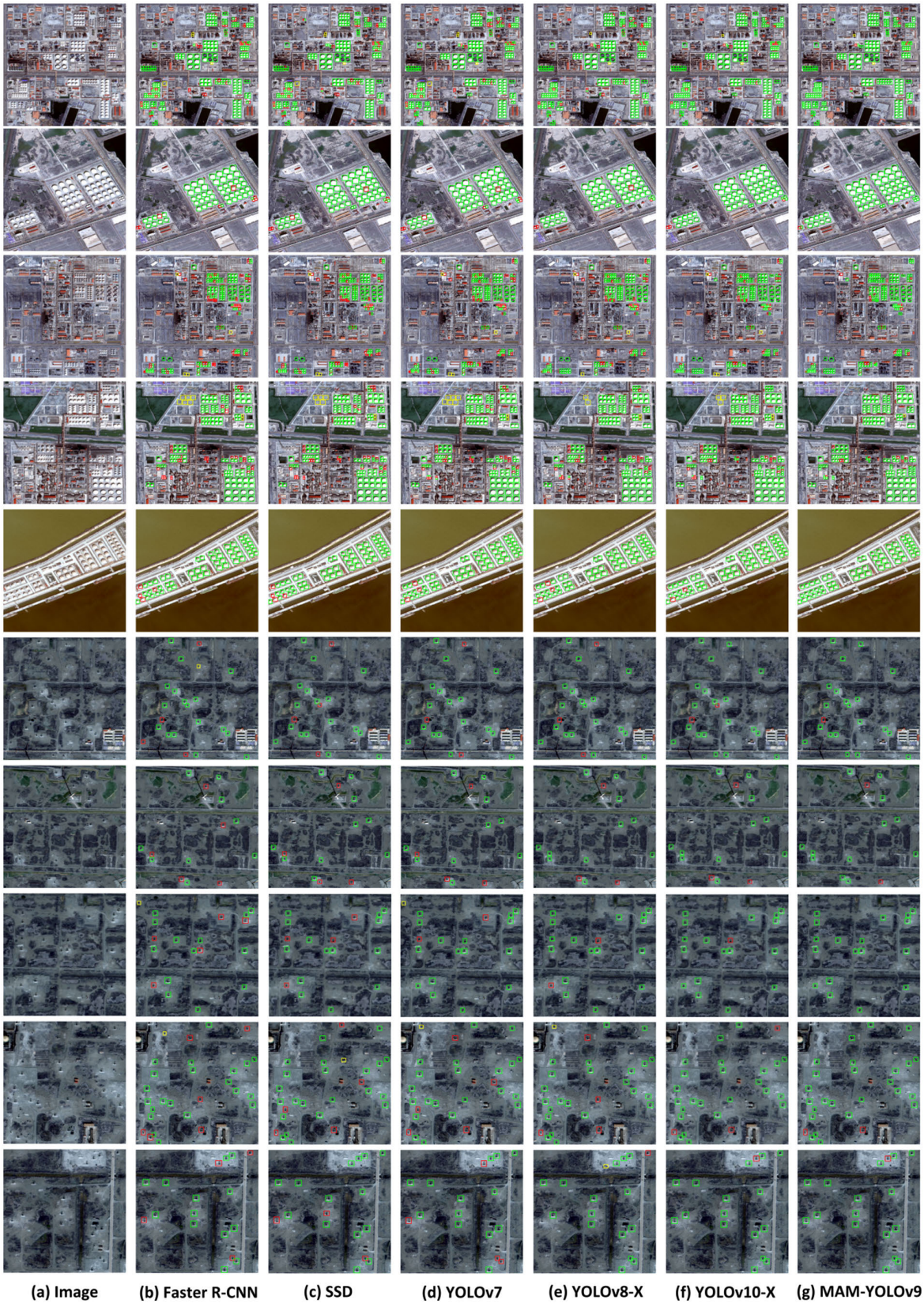


Fig. 9. Compare our proposed method with other state-of-the-art methods and visualize the detection results. The green box represents the correct detection result, the yellow box represents the incorrect detection result, and the red box represents the missing target. (a) Image. (b) Faster R-CNN. (c) SSD. (d) YOLOv7. (e) Yolov8-x. (f) Yolov10-x. (g) Ours MAM-YOLOv9.

with the predicted results of the model. From Table IV, it can be see that the number of predictions, positive detections, false detections, and recall rate are negatively correlated with window size, while accuracy is approximately positively cor-

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TABLE III
COMPARISON OF DETECTION RESULTS USING DIFFERENT METHODS

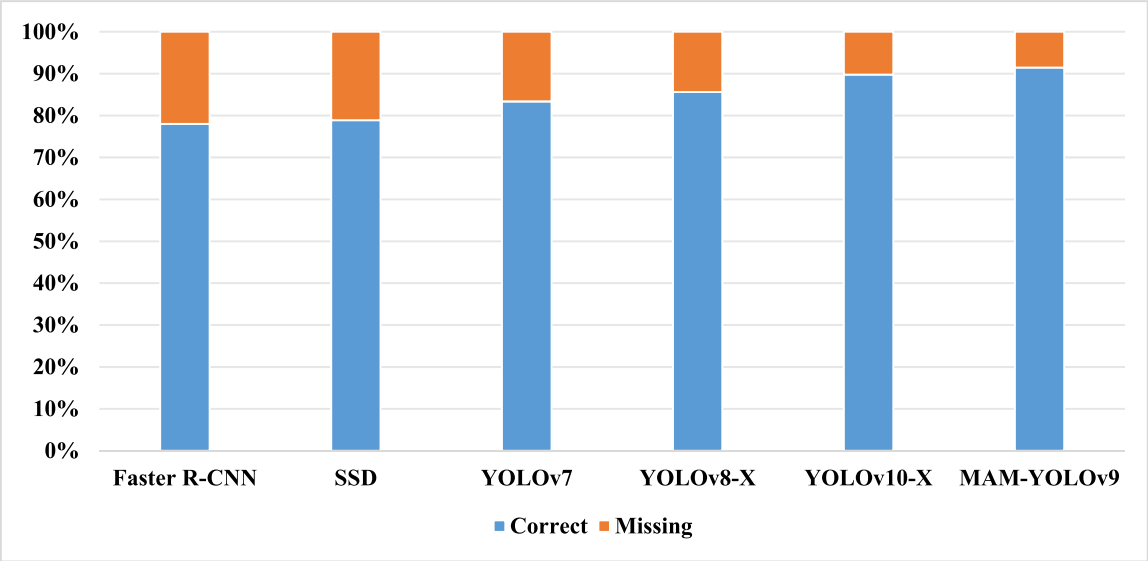


TABLE IV
STATISTICAL RESULTS OF DIFFERENT WINDOW SIZES

Window size	Actual	Detected	Correct(nTP)	Error(nFP)	Missing(nFN)	Precision	Recall
300×300	2698	2766	2565	201	133	0.927	0.951
480×480		2702	2544	158	154	0.942	0.943
640×640		**2579**	**2510**	**69**	**188**	**0.973**	**0.930**
720×720		2434	2341	93	357	0.962	0.868

related with window size. In the practical application scenario of methane emission facility detection, it is necessary to simultaneously consider both accuracy and recall indicators. Finally, the window size is set to 640 × 640 pixels. After detecting 2449 oil storage tanks and ensuring a recall rate of 92.6% for the model, the accuracy reached 96.8%. In the subsequent experiments, we will choose a window size of 640 × 640 pixels.

E. Ablation Study

Further verify the interrelationships between the designed modules and the impact of individual modules on algorithm performance. We are currently conducting ablation experiments, as shown in Table V. The performance of the proposed MAM-YOLOv9 is evaluated through thorough ablation studies, and it is found that compared with the original YOLOv9 model, the series of modules proposed in this article significantly improved recognition accuracy and precise classification. Each module contributes to improving the performance of the MAM-YOLOv9. The introduction of SimAM, CGA, and DyConv increased the mAP index by 2.17%, 1.03%, and 2.24%, respectively. The combination of modules can also improve the performance of the model; although the ability to improve performance varies, they all play a role in improving the model. Integrating these three modules together into the model achieved SOTA performance, making mAP@0.5 and mAP@0.5–0.9 increased by 5.37% and 3.07%, respectively.

F. Results

Carranza et al. [55] focused on developing the Vista-LA database for methane emission infrastructure in Los Angeles; Rafiq et al. [56] plotted the California Methane Emissions Infrastructure Database (Vista-CA); Omara et al. [57] developed a global scale Oil Infrastructure Geospatial Database (OGIM-v1.0) based on the acquisition, management, and integration of public geospatial datasets. However, the part of these methane emission infrastructure datasets related to oil storage tanks is relatively vague, only including nonpoint source-level data such as oil refineries and oil terminals, while point source-level oil storage tank data are still blank. Although the part related to oil wells is more detailed, including point source-level oil well data, it lacks relevant data from Asian countries such as China, and this gap urgently needs to be filled.

In this study, we used the MAM-YOLOv9 method to identify methane emission facilities in the Shengli Oilfield area and constructed a vector database of methane emission infrastructure named MEFD-v1.0. The image size of the Shengli Oilfield research area is 248 976 × 197 688 pixels, approximately 27239.64km^2. Through the MAM-YOLOv9 method and the aforementioned high-resolution image processing techniques,

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TABLE V
RESULTS OF ABLATION EXPERIMENT

Model	Yolov9	SimAM	CGA	DyConv	Precision	Recall	mAP@0.5	mAP@0.5~0.95
1	√				0.9114	0.9042	0.9390	0.7742
2	√	√			0.9224	0.9204	0.9594	0.7888
3	√		√		0.9129	0.9154	0.9487	0.7861
4	√			√	0.9280	0.9327	0.9600	0.7883
5	√	√	√		0.9478	0.9367	0.9728	0.7881
6	√	√		√	0.9563	0.9475	0.9796	0.7931
7	√		√	√	0.9693	0.9564	0.9849	0.7955
8	√	√	√	√	**0.9734**	**0.9666**	**0.9894**	**0.7980**

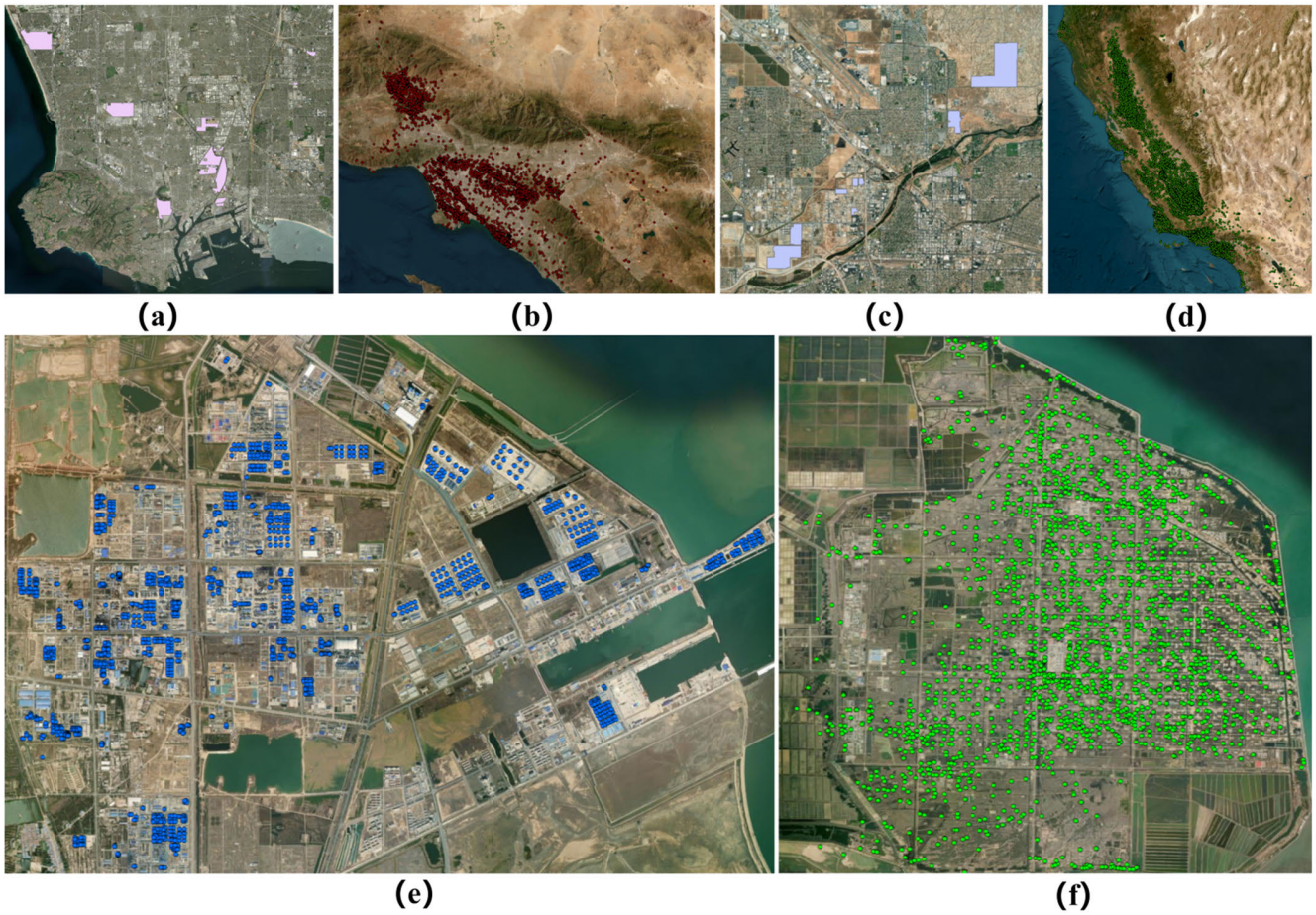


Fig. 10. Comparison between our proposed methane emission facility database (MEFD-v1.0) and Vista-LA and Vista-CA. (a) Vista-LA petroleum refineries. (b) Vista-CA refineries. (c) Vista-LA oil and gas wells. (d) Vista-CA oil and gas wells. (e) Oil tanks in verification area. (f) Oil wells in verification area.

this experiment collected 24 026 oil tanks and 14 958 oil well percussion machines on high-resolution images in the Shengli Oilfield area. To further validate the accuracy of the methane emission facility vector database, this article selected some areas as shown in Fig. 10 for verification. According to the results of image verification, there are 2190 oil tanks and 2342 oil wells in the area. Through MAM-YOLOv9, a total of 2023 oil tanks and 2192 oil wells were correctly detected, with accuracy rates of 92.37% and 93.60%, respectively. Due to different sizes of oil tanks and different orientations and types of oil wells, MAM-YOLOv9 still has missed detections, but the overall positive detection rate exceeds 90%. It can be considered that the methane emission facility vector database basically covers most of the methane emission facilities in the area.

By following the above process to detect and vectorize the methane emission facilities in the research area, a vector database of methane emission facilities in Shengli Oilfield can be obtained, as shown in Fig. 11. By comparison, we can see that OGIM only has a small amount of oil tank and well data in the eastern region of Asia, and there is no data distribution in China. Based on the methane emission facility database compiled in this article, we can clearly understand the quantity and distribution of oil tanks and wells in some areas of Shengli Oilfield. On this basis, relevant departments can carry out more detailed control over the densely distributed

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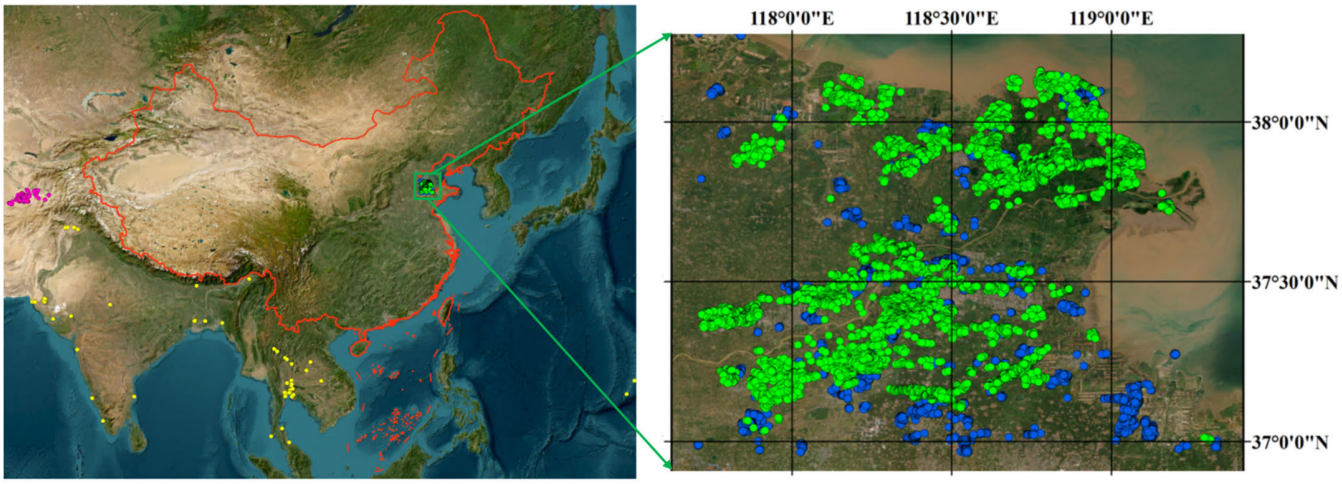


Fig. 11. Comparison between our proposed methane emission facility database (MEFD-v1.0) and OGIM-v1.0. The pink dots represent the oil and gas well data in OGIM. The yellow dots represent the petroleum refinery data in OGIM. The green dots represent the oil well data in MEFD-v1.0. The blue dots represent the oil tank data in MEFD-v1.0.

areas of oil wells and oil tanks, and then guide environmental work such as methane leakage and methane emission reduction. It is worth noting that our work fills the gap in this field's database and contributes to global methane and carbon monitoring.

V. Conclusion

This article focuses on methane emission facilities in oil fields, with Shengli Oilfield as the research area, and proposes an improved MAM-YOLOv9 method to identify methane emission facilities and construct a vector database. In terms of detection algorithms, the improved MAM-YOLOv9 method integrates SimAM into the YOLOv9 network and captures background noise information around small targets, which is beneficial for distinguishing targets from their surrounding environment; On the basis of the original CGA module, improvements were made to increase the network depth, thereby improving the computational efficiency of the model without introducing any additional parameters; replace DyConv with Conv to improve computational efficiency and object detection accuracy. The experimental results show that the MAM-YOLOv9 model outperforms current mainstream object recognition algorithms in terms of accuracy and inference time. Integrating these three modules together into the model achieved SOTA performance, making mAP@0.5 and mAP@0.5 0.9 increased by 5.37% and 3.07%, respectively. Compared with other advanced object detection algorithms, this method has achieved excellent detection results. Through the MAM-YOLOv9 method, we are able to identify and vectorize methane emission facilities in some regions of China, filling a gap in the industry. In addition, this article also created and released a methane emission facility sample dataset containing 2000 sample data, which can help train more complex and accurate deep learning models and improve model performance and generalization ability.

However, there are still some challenges that need to be addressed, such as improving the generality of the model across different datasets and environments. More research is needed to investigate the effectiveness conditions and environmental factors of our proposed model in high-resolution imaging. However, in this article, the combination of deep learning technology and China's high-resolution satellite series has successfully achieved the detection and identification of methane emission facilities. This unique method not only injects a new concept into the field of methane emission monitoring but also cleverly integrates satellite technology with cutting-edge data processing technology, thereby achieving more accurate and efficient emission source tracking results. This innovative work provides strong support for progress in the fields of environmental protection and climate change, paving the way for the realization of a sustainable future.

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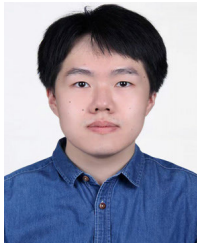


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